

Rules and Regulations for **Mini Sumo** Category

1. About Competition

The Mini Sumo category is a robotics competition where young participants design and program autonomous robots to compete in sumo-style matches. The category encourages creativity, technical knowledge, teamwork, and problem-solving skills while promoting fair play and innovation. It provides students with an exciting opportunity to explore robotics, engineering, and technology in a competitive environment.

2. Robot Specification

1. Materials and Equipment:

- Participants may use any materials of their choice for robot construction, provided they are safe and comply with competition rules.
- The Mini Sumo Robot must be built by participants. Robots **must not be purchased** from any company. If a robot is found to have been purchased, the team will be **disqualified** from the competition.
- Participants may use 3D parts for their robots.

2. Size and Weight:

- The robot must not exceed a 10 cm × 10 cm boundary.
- The robot must not exceed a maximum weight of 500 grams.
- No height limit.

3. Power and Signal Communication :

- The robot must work autonomously by power source.
- Wi-Fi and Bluetooth must be turned off during the match. The program must be uploaded(via USB) before the match.
- Participants may use an IR sensor to start or stop the robot before the round.

3. Match Regulations

1. Start:

- In total there will be 3 rounds per match. Both teams proceed to the ring to position their robots on the competition surface in accordance with the match setup procedure.

2. Match Duration:

- Each round will last a maximum of 3 minutes.
- If the robots fail to win within 3 minutes, then this round will be played again.

3. Victory Conditions:

- The robot wins if it pushes another robot away from the dohyo.
- If half of the robot goes beyond the white part of the dohyo, then this robot will lose and the point will be counted to the other robot.
- If both robots go beyond the white part of the dohyo the round will be counted as a draw and will be played again.

4. Match Procedure:

- The round will begin with a signal from the referee.
- Referees may stop the round if there are some problems.
- When the round begins, the robot shall observe a 5-second countdown after activation before initiating any movement.
- Connecting to the robot via Bluetooth or Wi-Fi is prohibited during matches. If anyone connects to the robot via Bluetooth or Wi-Fi, the team will be disqualified.
- The robot is permitted to deploy or unfold components after movement begins; however, it must remain a single, unified structure at all times and is not allowed to detach into separate parts.
- The robots must clearly display an identification number assigned by the organizers on the exterior casing in a clearly visible position.

5. Repairs and Maintenance

- Breaks between rounds are not permitted unless an unexpected or exceptional circumstance occurs.
- If a robot experiences a failure during a match, the referee may grant a 2-minute repair period. All repairs must be carried out under the supervision of an assistant referee to ensure that only approved components are used and no unauthorized modifications are made.
- If the robot is not repaired within 2 minutes, it will lose and the point will go its opponent
- A competitor may request a stoppage of the match in the event of personal injury or significant robot damage that prevents the match from continuing. A break will be granted if the referee determines it is necessary and justified.

4. Sportsmanship and Fair Play

1. Sportsmanship:

- Participants are expected to demonstrate fairness, respect, and good sportsmanship throughout the competition.
- Unsportsmanlike conduct may lead to disqualification. The Referee reserves the right to disqualify any supervisor, participant, or robot found to be acting unethically or in violation of sportsmanship standards. All decisions made by the jury are final and not subject to appeal.

2. Inspections:

- Before the competition, the robots will be inspected to ensure they comply with rules
- Referees may ask you about the robot to ensure that the student built the robot by itself. Referees may ask you before or after the competition.

3. Participant behaviour:

- Participants are not allowed to touch the robot during the match.
- Having communication between participants and the referees is strictly prohibited
- During the game, 1 participant and 1 assistant must be present during the game. Other participants cannot be with them.

4. Maintenance:

- Participants may repair the robot before the match starts.
- Mentor and Assistant can help you and even give you advice during the game.
- If there's a problem with the robot during the match, the referee will give you 2 minutes to fix it. A mentor or assistant can help you.

5. Authority of the Judges:

- The Judge has the authority to supervise all matches and ensure that the competition rules are followed correctly by all participants.
- The Judge has the power to make final decisions regarding victories, disqualifications, rule violations, and match results during the competition.
- The Judge is responsible for maintaining fairness, safety, and discipline throughout the event and may stop a match if necessary.
- The Judge may provide warnings or penalties to participants who do not follow the official regulations and competition procedures.

5. Safety Rules

1. Safety of Robot:
 - The robot must not have any parts that could harm the participant or robot.
 - The referee will check your robot for safety, and if it's accepted, you'll be able to participate. If not, you'll have time to fix it.
 - Participants are forbidden from using any device that can throw or discharge liquids, dust, gas, or other substances at an opponent.
2. Approval:
 - All robots will be checked for safety to ensure compliance with the rules.
 - If any safety issues are found with the robot, participants will have time to fix it or it will be disqualified.

6. Dohyo Specification

1. The ring consists of a circular platform with a diameter of 77 cm(770 mm).
2. The surface is black with a 2.5(25mm) cm white boundary line.
3. The robot **must not** damage the dohyo.

7. Disqualification

1. The robot will be disqualified if the robot does not comply with the rules.
2. The robot will be disqualified if it damages the robot or participant.
3. If a participant shouts or goes against the referee.
4. If a participant will disrespect other teams.
5. If the robot won't be fixed during the match.
6. The robot must not be physically modified or have its code changed after inspection.
7. If the participant won't come after the call-up.
8. If you are called to play and you don't come in 5 minutes the robot will lose the match